UBUNTU 24.04 LTS

Complete Installation Case Study & Technical Reference

From Bootable USB Creation to a Fully Operational Linux System

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Abstract

This document presents a comprehensive, step-by-step technical case study of installing Ubuntu 24.04 LTS (Noble Numbat) on a new personal computer. The installation involved creating a bootable USB drive using Rufus on Windows, configuring BIOS/UEFI settings, navigating the Ubuntu interactive installer, and resolving multiple real-world boot failures encountered during the process. The document covers: pre-installation requirements, tool selection rationale, disk partitioning strategy, bootloader (GRUB) behaviour, post-installation verification procedures, and a structured analysis of every error encountered alongside the applied solution. All terminal commands, system outputs, and configuration parameters are documented verbatim for reproducibility. This record serves as both a reference guide for system administrators and a research artifact illustrating the practical challenges of bare-metal Linux deployment.

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1. Introduction

Installing a Linux-based operating system on a personal computer from scratch requires a thorough understanding of firmware interfaces (BIOS/UEFI), disk partitioning schemes (MBR vs GPT), bootloader configuration (GRUB), and filesystem conventions. This case study documents a real-world installation journey of Ubuntu 24.04 LTS on a new PC, capturing each decision point, every command executed, all errors encountered, and the exact resolution strategies employed.

Unlike controlled laboratory installations, this process encountered authentic issues including bootloader conflicts between Windows and Ubuntu, an initramfs emergency shell failure, and a USB boot priority mismatch. Each incident is catalogued in Section 11 with its root cause and resolution. The goal of this document is to serve as a reproducible, academically rigorous record that can guide future installations and inform research into OS deployment challenges on consumer hardware.

2. Prerequisites and Hardware Specifications

The following hardware, software, and environmental prerequisites were confirmed before beginning the installation process:

Host Machine	New PC with NVMe SSD (nvme0n1)
Bootable Media	USB Pendrive (minimum 8 GB)
Tool Used	Rufus (Windows) for USB creation
Ubuntu Version	Ubuntu 24.04 LTS (Noble Numbat)
ISO File	ubuntu-24.04-desktop-amd64.iso (~5.7 GB)
Partition Scheme	GPT (GUID Partition Table)
Firmware	UEFI (confirmed via msinfo32 → BIOS Mode: UEFI)
Prior OS	Windows (completely removed during installation)
Installation Type	Full Disk Erase — Ubuntu Only (no dual boot)

2.1 Software Tools Used

- Rufus v4.x (Windows utility for creating bootable USB drives)
- Ubuntu 24.04 LTS ISO image (ubuntu-24.04-desktop-amd64.iso)
- Windows System Information tool (msinfo32) for BIOS mode detection
- Ubuntu Live Environment (for emergency GRUB repair)

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2.2 Network Requirements

- Active internet connection recommended during installation (for updates and driver downloads)
- Minimum 10 GB free disk space (recommended: full disk)

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3. Phase 1 — Downloading the Ubuntu ISO

The first step in any OS installation is obtaining the official installation media. For Ubuntu, this is distributed as an ISO 9660 disk image.

3.1 Official Download Source

The Ubuntu ISO was downloaded from the official Canonical website:

```
URL: https://ubuntu.com/download/desktop
File: ubuntu-24.04-desktop-amd64.iso
Size: ~5.7 GB
Architecture: amd64 (x86_64, 64-bit)
```

3.2 Why Ubuntu 24.04 LTS?

LTS stands for Long Term Support. Ubuntu 24.04 LTS (Noble Numbat) receives security patches and official updates for five years (until April 2029). This makes it the preferred choice for research machines, production servers, and personal workstations where stability and longevity are valued over cutting-edge but potentially unstable features found in non-LTS releases.

> **ℹ Version Selection Rationale**
> Ubuntu releases a new LTS version every two years (e.g., 20.04, 22.04, 24.04). For any serious deployment, always choose the latest LTS release for maximum hardware support and the longest maintenance window.

3.3 File Integrity

After download, the ISO file should ideally be verified against the official SHA256 checksum provided by Canonical. This ensures the downloaded file has not been corrupted or tampered with:

```
# On Linux/macOS:
sha256sum ubuntu-24.04-desktop-amd64.iso

# On Windows (PowerShell):
Get-FileHash ubuntu-24.04-desktop-amd64.iso -Algorithm SHA256
```

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4. Phase 2 — Determining Partition Scheme (GPT vs MBR)

Before creating a bootable USB, it is essential to determine the correct partition scheme. Using the wrong scheme will result in the system failing to boot from the USB drive.

4.1 Understanding the Difference

There are two major disk partition table formats:

Feature	MBR (Master Boot Record)	GPT (GUID Partition Table)
Firmware	BIOS (Legacy)	UEFI
Max Disk Size	2 TB	9.4 ZB (virtually unlimited)
Max Partitions	4 primary	128 (on most OSes)
Introduced	1983	2000s
Used on	Old/Legacy PCs	Modern PCs (post-2012)
Boot Partition	MBR at start of disk	EFI System Partition (ESP)

4.2 Detection Method Used

To determine whether the current system uses UEFI or Legacy BIOS — which dictates whether GPT or MBR should be used — the Windows System Information tool was used:

```
Press: Windows + R
Type:  msinfo32
Press: Enter

Look for the field: BIOS Mode
  → If it reads 'UEFI'   → Use GPT in Rufus
  → If it reads 'Legacy' → Use MBR in Rufus
```

4.3 Decision Made

✓ GPT Selected

The host machine confirmed UEFI firmware via msinfo32. Therefore, GPT was selected as the partition scheme in Rufus. This is correct for all modern (post-2015) computers.

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5. Phase 3 — Creating a Bootable USB with Rufus

Rufus is a free, open-source Windows utility for creating bootable USB flash drives. It is significantly faster than competing tools (e.g., UNetbootin, Etcher) and supports both ISO and DD image modes.

5.1 Rufus Configuration (Ubuntu)

The following exact configuration was applied in Rufus to create the Ubuntu bootable USB:

```
Device           → [Your USB Drive - e.g., SanDisk 16GB]
Boot selection   → ubuntu-24.04-desktop-amd64.iso
Image option     → Standard Windows Installation  [auto-detected as ISO
mode]
Partition scheme → GPT
Target system    → UEFI (non-CSM)  [auto-set]
File system      → FAT32            [auto-set]
Cluster size     → 4096 bytes       [default]
Volume label     → Ubuntu 24.04 LTS [auto-set from ISO]

Click: START
Prompt: Write in ISO Image mode (Recommended) → OK
WARNING: All data on USB will be destroyed → OK
```

5.2 Process Duration

Writing the bootable USB took approximately 8–15 minutes depending on USB speed (USB 3.0 recommended for faster writes). Rufus displays a progress bar and shows READY when complete.

> ⚠ **Critical Warning**
> Never remove the USB drive while Rufus is writing. Interrupting the write process corrupts the drive and makes it non-bootable.

5.3 Creating a Windows 11 Bootable USB with Rufus (Interim Step)

During the installation process, a Windows 11 bootable USB was also created (before the final Ubuntu-only decision was made). The Rufus configuration for Windows 11 differed slightly:

```
Device           → [USB Pendrive]
Boot selection   → Win11_24H2_EnglishInternational_x64.iso
Image option     → Standard Windows Installation
Partition scheme → GPT
Target system    → UEFI (non-CSM)
```

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```
Optional (Rufus dialog after clicking START):
  ☑ Remove requirement for 4GB+ RAM, Secure Boot and TPM 2.0
  ☑ Remove requirement for Microsoft account
→ Click OK
```

Windows 11 ISO source:

```
https://www.microsoft.com/software-download/windows11
```


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6. Phase 4 — Booting from USB (BIOS/UEFI Boot Menu)

With the bootable USB prepared, the next step is to configure the PC to boot from the USB drive instead of its internal hard disk.

6.1 Entering the Boot Menu

On most modern PCs, a one-time boot menu can be accessed by pressing a specific key immediately after powering on the machine. The key varies by manufacturer:

Manufacturer	Boot Menu Key
Lenovo (ThinkBook, ThinkPad)	F12
Dell	F12
HP	F9 or ESC
ASUS	F8 or ESC
Acer	F12
MSI	F11
Generic / Unknown	F12 / F2 / ESC / DEL (try in order)

6.2 Boot Menu Navigation

Once inside the boot menu, the USB drive appears under one of these labels depending on the system:

- USB: SanDisk Ultra (or your USB brand name)
- UEFI: SanDisk ...
- Generic Storage (UEFI)

Select the USB entry and press Enter. The system will begin loading the Ubuntu installer from the USB drive.

6.3 GRUB Boot Screen Options

After the system boots from USB, the GRUB bootloader on the USB presents the following menu:

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```
*Try or Install Ubuntu
 Ubuntu (safe graphics)
 OEM install (for manufacturers)
 Boot from next volume
 UEFI Firmware Settings
```

During the initial installation attempt, 'Try or Install Ubuntu' was selected. In a subsequent repair attempt (after encountering boot issues), 'Ubuntu (safe graphics)' was used to bypass a graphics driver conflict.

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7. Phase 5 — Interacting with the Ubuntu Installer

The Ubuntu 24.04 installer uses a new Qt-based graphical interface (replacing the older Ubiquity installer). The following screens were navigated in order:

7.1 Installer Step-by-Step Walkthrough

Step 1: Language Selection

Language: English (United States)

Step 2: Accessibility

Default settings retained. No assistive technologies required.

Step 3: Keyboard Layout

Layout: English (US) — Default. Detected automatically.

Step 4: Network Connection

Connected to available Wi-Fi/Ethernet. Not strictly required but recommended for downloading updates during installation.

Step 5: Installation Type

```
Selected: Interactive Installation
  (Guided, wizard-based — recommended for most users)

Alternative not selected: Automated Installation
  (Uses a YAML config file — for enterprise/scripted deployments)
```

Step 6: Application Selection

```
Selected: Default Selection
  (Includes: Firefox, LibreOffice, Thunderbird, basic utilities)

Also selected:
  ☑ Install third-party software for graphics and Wi-Fi hardware
  ☑ Download and install support for additional media formats
```

> **ℹ Why Third-Party Drivers Matter**
>
> Selecting this option installs proprietary firmware for Wi-Fi adapters (Broadcom, Intel), NVIDIA graphics drivers, and media codecs. Skipping this often results in a system with no Wi-Fi connectivity or broken video playback immediately after installation.

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Step 7: Disk Configuration (CRITICAL)

This is the most consequential step of the installation. The wrong choice here can result in data loss or an unbootable system.

```
Disk: nvme0n1  (NVMe SSD — the internal storage of the PC)

Installation type selected:
  ● Erase disk and install Ubuntu
     → Deletes ALL partitions on nvme0n1
     → Creates fresh GPT partition table
     → Creates EFI System Partition (/boot/efi)
     → Creates root partition (/) with ext4 filesystem
     → Creates swap partition

Advanced features:
  ○ Use LVM with new Ubuntu installation   [NOT selected]
  ○ Encrypt the new Ubuntu installation    [NOT selected]
```

The resulting partition layout after selection:

```
nvme0n1
├── nvme0n1p1   512 MB    EFI System Partition (FAT32)   → /boot/efi
├── nvme0n1p2   1 GB      Linux filesystem (ext4)        → /boot
└── nvme0n1p3   [remaining space]  Linux filesystem (ext4) → /

Swap: Managed via swapfile (not a dedicated partition in 24.04)
```

> ⚠ **Data Loss Warning**
>
> Selecting 'Erase disk and install Ubuntu' permanently destroys all data on the selected disk. In this case, this was intentional — the goal was to completely remove Windows and install Ubuntu as the sole operating system.

Step 8: User Account Creation

```
Full name:      [User-defined]
Computer name:  [User-defined hostname]
Username:       [User-defined]
Password:       [User-defined — minimum 6 characters]

Login option:
  ● Require my password to log in  (selected)
  ○ Log in automatically           (not selected)
```

Step 9: Timezone Configuration

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```
Region:   Asia
Zone:     Kolkata  (UTC+5:30 — India Standard Time)
```

Step 10: Installation

After confirmation, the installer began writing to disk. The installation process included:

1. Formatting partitions (ext4 on root, FAT32 on EFI)
2. Extracting base system files from the squashfs image on the USB
3. Installing kernel and bootloader (GRUB 2)
4. Generating initramfs (initial RAM filesystem)
5. Configuring locale, timezone, and user accounts
6. Installing third-party drivers (if selected)

Approximate duration: 10–20 minutes depending on hardware speed.

Step 11: Restart

```
After installation completes:
  → Click: Restart Now
  → Prompt: 'Please remove the installation medium, then press ENTER'
  → Action: Remove USB pendrive
  → Press: ENTER
```

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8. Phase 6 — Post-Installation Verification

After the first successful installation and login, the following terminal commands were used to verify system integrity and confirm Windows had been completely removed:

8.1 Verify Root Filesystem

```
$ ls /
bin  boot  dev  etc  home  lib  lib64  lost+found  media  mnt
opt  proc  root  run  sbin  snap  srv  sys  tmp  usr  var
```

Presence of standard Linux directory hierarchy (FHS-compliant) confirms Ubuntu is installed correctly.

8.2 Verify Disk Partitions

```
$ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
nvme0n1     259:0    0 [DISK SIZE] 0 disk
├─nvme0n1p1 259:1    0   512M  0 part /boot/efi
├─nvme0n1p2 259:2    0     1G  0 part /boot
└─nvme0n1p3 259:3    0  [remaining] 0 part /
```

Key observations from lsblk output:

- Only Linux partition types present (no ntfs = Windows is gone)
- EFI partition mounted at /boot/efi
- Root partition mounted at /
- No dual-boot entries

8.3 Verify OS Information

```
$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04 LTS (Noble Numbat)"
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
```

8.4 Install System Info Tool (Optional)

```
$ sudo apt install neofetch
$ neofetch
```

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```
Displays: OS, kernel version, uptime, CPU, GPU, RAM, disk usage
```

8.5 Check GRUB Configuration

```
$ sudo cat /boot/grub/grub.cfg | head -30
# Confirms GRUB is installed and configured for Ubuntu boot

$ sudo grub-install --version
grub-install (GRUB) 2.12
```

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9. Phase 7 — Creating a Windows 11 Bootable USB (Interim Step)

At an intermediate point in the installation journey, a decision was made to temporarily test Windows 11. The Ubuntu-bootable USB was reformatted to create a Windows 11 bootable drive. This section documents that process for completeness.

Note: The USB does NOT need to be manually formatted before using Rufus. Rufus automatically wipes and reformats the drive during the image-writing process.

9.1 Rufus Configuration for Windows 11

```
Step 1: Download Windows 11 ISO
  URL: https://www.microsoft.com/software-download/windows11
  Select: Windows 11 (multi-edition ISO, 64-bit)
  Select language: English (International)
  File size: ~6.0 GB

Step 2: Open Rufus and set:
  Device           → [USB Pendrive]
  Boot selection   → Win11_24H2_EnglishInternational_x64.iso
  Image option     → Standard Windows Installation
  Partition scheme → GPT
  Target system    → UEFI (non-CSM)

Step 3: Click START
  Dialog appears: Windows User Experience
  ☑ Remove requirement for 4GB+ RAM, Secure Boot and TPM 2.0
  → Click OK

Step 4: Rufus formats USB and writes Windows image (~10 min)
Step 5: When status shows READY — USB is ready
```

> **ℹ Note on Windows TPM Bypass**
>
> Modern Windows 11 installations require TPM 2.0, Secure Boot, and 4GB RAM. Rufus provides an option to bypass these requirements, which is useful for testing on older or non-compliant hardware. This is an offline bypass only — Windows activation and updates may still require compliant hardware.

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10. Phase 8 — Re-installing Ubuntu (Full Disk Wipe)

After the Windows 11 test, the decision was made to fully commit to Ubuntu as the only operating system. The USB was reformatted back to Ubuntu (using Rufus as described in Section 5), and a clean installation was performed.

This re-installation followed all steps from Phase 4 through Phase 6. The key difference from the earlier installation attempts was:

- Strictly selecting 'Erase disk and install Ubuntu' (not 'Install alongside Windows')
- Ensuring no residual Windows Boot Manager entry remained in the UEFI NVRAM
- Selecting 'Ubuntu (safe graphics)' from GRUB on the USB to avoid graphics driver conflicts during installation
- Properly removing the USB when prompted before the first reboot

After this clean installation, the system booted directly into Ubuntu without any GRUB dual-boot menu, Windows Boot Manager interference, or initramfs errors.

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11. Problems Faced and Solutions Applied

This section is the core diagnostic record of the installation. All errors encountered during the process are documented here with their root cause analysis and the exact solution applied.

Problem 1: System Still Booted Into Windows After Ubuntu Installation

✕ Symptom

After completing Ubuntu installation and restarting, the system booted directly into Windows. Ubuntu was not visible in the boot sequence.

Root Cause

The UEFI NVRAM boot order still had 'Windows Boot Manager' as the highest priority entry. Ubuntu's GRUB bootloader was installed to the EFI System Partition, but the UEFI firmware ignored it because Windows Boot Manager had higher priority.

Diagnostic Steps

```
1. Restart PC → Press F12 → Open Boot Menu
2. Observe boot entries:
   - Windows Boot Manager   ← Present (high priority)
   - Ubuntu                 ← Missing or lower priority
```

Solutions Attempted

7. Boot via F12 one-time boot menu → manually select Ubuntu each time (temporary workaround)
8. Enter BIOS → Boot Priority → Move Ubuntu above Windows Boot Manager → Save
9. From within Ubuntu terminal, run GRUB update commands:

```
$ sudo grub-install /dev/nvme0n1
$ sudo update-grub
$ sudo reboot
```

✓ Final Resolution

The cleanest resolution was a fresh re-installation using 'Erase disk and install Ubuntu', which deleted the Windows Boot Manager EFI entry entirely and registered Ubuntu's GRUB as the sole UEFI boot entry.

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Problem 2: initramfs BusyBox Emergency Shell on Boot

✕ Symptom

After an installation attempt, the system failed to boot and dropped into a BusyBox v1.36.1 (initramfs) shell with an error similar to: 'gave up waiting for root file system device' or 'Unable to find a medium containing a live file system'.

Error Message Observed

```
BusyBox v1.36.1 (Ubuntu 1:1.36.1-6ubuntu3) built-in shell (ash)
Enter 'help' for a list of built-in commands.

(initramfs) _
```

Root Cause

The initramfs (initial RAM filesystem) failure occurs during the early boot stage, before the main root filesystem is mounted. Causes identified in this case:

- Incomplete installation: The previous installation was interrupted or did not write the bootloader correctly
- Partition conflict: Windows Boot Manager's presence caused GRUB to write to a conflicting location on the EFI System Partition
- Corrupted GRUB installation: The grub-install command did not properly register the EFI boot entry

Attempted Recovery Commands

```
# From the initramfs shell, basic diagnostics:
(initramfs) ls /dev/
(initramfs) ls /dev/nvme*

# Attempt filesystem check:
(initramfs) fsck /dev/nvme0n1p3

# Exit and attempt normal boot:
(initramfs) exit

# Reboot:
(initramfs) reboot
```

GRUB Repair via Live USB (Attempted)

```
# Boot from Ubuntu USB → Choose 'Try Ubuntu'
# Open Terminal (Ctrl+Alt+T)

# Step 1: Identify partitions
$ lsblk
```

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```
# Step 2: Mount root partition
$ sudo mount /dev/nvme0n1p3 /mnt

# Step 3: Mount EFI partition
$ sudo mount /dev/nvme0n1p1 /mnt/boot/efi

# Step 4: Bind system directories
$ sudo mount --bind /dev /mnt/dev
$ sudo mount --bind /proc /mnt/proc
$ sudo mount --bind /sys /mnt/sys

# Step 5: Chroot into installed system
$ sudo chroot /mnt

# Step 6: Reinstall GRUB
$ grub-install --target=x86_64-efi --efi-directory=/boot/efi --
bootloader-id=ubuntu
$ update-grub

# Step 7: Exit and reboot
$ exit
$ sudo reboot
```

✓ Final Resolution

A clean re-installation using 'Erase disk and install Ubuntu' resolved the initramfs error permanently. The fresh install wrote a clean partition table, correctly generated initramfs, and installed GRUB to the EFI System Partition without conflicts.

Problem 3: Ubuntu Not Visible in UEFI Boot Entries

✕ Symptom

When entering BIOS Boot Manager or UEFI Firmware Settings, only 'Windows Boot Manager' appeared. There was no 'Ubuntu' or 'GRUB' entry. This meant even setting boot priority was not possible.

Root Cause

Ubuntu's GRUB EFI boot entry (shimx64.efi / grubx64.efi) was not registered in the UEFI NVRAM. This happens when:

- The grub-install process fails silently
- The EFI System Partition is not correctly mounted during installation
- Secure Boot rejects unsigned bootloaders (if enabled)

Diagnostic Command

```
# Check UEFI boot entries (from Linux Live environment):
```

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```
$ sudo efibootmgr -v

# Expected output (after correct install):
BootCurrent: 0003
BootOrder: 0003,0002
Boot0002* Windows Boot Manager
Boot0003* ubuntu

# Problem output (what was seen):
BootCurrent: 0002
BootOrder: 0002
Boot0002* Windows Boot Manager
```

Manual UEFI Entry Creation (Advanced)

```
# If needed, manually add Ubuntu EFI entry:
$ sudo efibootmgr --create \
    --disk /dev/nvme0n1 \
    --part 1 \
    --label 'ubuntu' \
    --loader '\EFI\ubuntu\shimx64.efi'
```

✓ Final Resolution

Clean re-installation with 'Erase disk and install Ubuntu' automatically deleted the Windows Boot Manager EFI entry and created a new Ubuntu UEFI boot entry. No manual efibootmgr intervention was needed.

Problem 4: USB Still Booting Into Linux After Re-purposing as Windows USB

✕ Symptom

After repurposing the Ubuntu USB as a Windows 11 USB using Rufus, the old Linux boot files were still present and sometimes loaded.

Root Cause

The USB drive had residual GRUB/Linux data in the MBR or EFI partition from the previous Ubuntu bootable image. Rufus replaces the filesystem but sometimes leaves residual low-level boot code.

Solution

```
Solution 1: Let Rufus fully format the drive before writing
  → Rufus automatically formats the USB when writing a new image.
  → No manual formatting needed.
```

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```
Solution 2 (if Rufus fails): Manually clean the USB using diskpart
(Windows)
  > diskpart
  DISKPART> list disk
  DISKPART> select disk [USB disk number]
  DISKPART> clean
  DISKPART> create partition primary
  DISKPART> format fs=fat32 quick
  DISKPART> assign
  DISKPART> exit

Then re-run Rufus to write the Windows 11 ISO.
```

Problem 5: Confusion About 'Same User' After Fresh Installation

✕ Symptom

After the clean Ubuntu installation and reboot, the login screen showed the same username that was created during setup. The user was uncertain whether the installation had succeeded or whether old Windows data still existed.

Explanation

This is expected behaviour and NOT an error. The Ubuntu installer creates a user account during setup. After installation and reboot, Ubuntu logs into that account. Seeing the same username is a sign the installation succeeded — the user account was created fresh during the installation process.

Verification Commands

```
# Confirm Ubuntu is installed (not running from USB):
$ df -h /
Filesystem      Size  Used Avail Use% Mounted on
/dev/nvme0n1p3  [X]G  [Y]G  [Z]G  [N]% /

# If running from USB, / would show: tmpfs or /dev/loop0

# Check for Windows filesystems (should be absent):
$ lsblk -f
# Look for: FSTYPE column
# Ubuntu: ext4 (root), vfat (EFI), swap
# Windows: ntfs (C:, D: drives) — should be absent
```

✓ Confirmed

Running lsblk and df confirmed that the system was booted from the NVMe SSD with ext4 root partition — a clean Ubuntu installation. No NTFS (Windows) partitions were present.

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11.6 Problem-Solution Summary Table

Problem	Root Cause	Solution Applied
System booted Windows after Ubuntu install	Windows Boot Manager had higher UEFI priority than Ubuntu GRUB	Fresh install with Erase disk option removed Windows EFI entry
initramfs BusyBox emergency shell	Incomplete/conflicted GRUB installation due to dual-boot partition conflict	Clean reinstall; GRUB repair via Live USB chroot as backup
Ubuntu not in UEFI boot entries	grub-install failed to register EFI entry in NVRAM	Fresh install auto-registered Ubuntu UEFI entry correctly
USB still booting old Linux after Rufus write	Residual GRUB/MBR data from previous Ubuntu USB image	Rufus auto-formats; diskpart clean used for stubborn cases
Same user shown after install — confusion	User account created during install is expected — not old data	Verified via lsblk (ext4 only, no ntfs) and df commands

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12. Key Terminal Commands Reference

The following is a consolidated reference of all commands used or recommended throughout the installation and debugging process.

12.1 System Identification

```
# View partition layout
lsblk
lsblk -f                    # Show filesystem types

# View mounted filesystems and disk usage
df -h

# View OS details
cat /etc/os-release
uname -a                    # Kernel version

# View UEFI boot entries
sudo efibootmgr -v

# View disk partition table
sudo fdisk -l /dev/nvme0n1
sudo gdisk -l /dev/nvme0n1  # For GPT
```

12.2 GRUB Bootloader Commands

```
# Install GRUB to EFI System Partition
sudo grub-install /dev/nvme0n1

# Targeted GRUB install for UEFI
sudo grub-install --target=x86_64-efi \
     --efi-directory=/boot/efi \
     --bootloader-id=ubuntu

# Regenerate GRUB configuration
sudo update-grub

# Check GRUB version
grub-install --version
```

12.3 Package Management

```
# Update package lists
sudo apt update

# Upgrade all packages
sudo apt upgrade
```

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```
# Install a specific package
sudo apt install neofetch
sudo apt install git curl wget build-essential

# Search for a package
apt search <package-name>

# Remove a package
sudo apt remove <package-name>
```

12.4 Filesystem and Mount Commands

```
# Mount a partition
sudo mount /dev/nvme0n1p3 /mnt

# Mount EFI partition
sudo mount /dev/nvme0n1p1 /mnt/boot/efi

# Bind mount system directories (for chroot)
sudo mount --bind /dev  /mnt/dev
sudo mount --bind /proc /mnt/proc
sudo mount --bind /sys  /mnt/sys

# Enter chroot environment
sudo chroot /mnt

# Filesystem integrity check
sudo fsck /dev/nvme0n1p3

# Unmount
sudo umount /mnt/dev /mnt/proc /mnt/sys /mnt/boot/efi /mnt
```

12.5 Power Management

```
# Reboot system
sudo reboot

# Power off system
sudo poweroff

# From initramfs emergency shell:
(initramfs) reboot
(initramfs) exit    # Attempt to continue boot
```

12.6 SHA256 Checksum Verification

```
# Linux/macOS:
sha256sum ubuntu-24.04-desktop-amd64.iso

# Windows PowerShell:
Get-FileHash ubuntu-24.04-desktop-amd64.iso -Algorithm SHA256
```

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```
# Compare with official hash from:
# https://releases.ubuntu.com/24.04/SHA256SUMS
```


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13. Conceptual Background

13.1 What is GRUB?

GRUB (GNU GRand Unified Bootloader) is the default bootloader for most Linux distributions, including Ubuntu. It is the first software that runs after the UEFI firmware hands over control. GRUB is responsible for:

- Loading the Linux kernel from the boot partition
- Passing kernel parameters (e.g., root device, quiet mode, splash screen)
- Displaying a boot menu when multiple OSes are installed (dual-boot)
- Loading the initial RAM disk (initrd/initramfs) that bootstraps the real OS

13.2 What is initramfs?

initramfs (initial RAM filesystem) is a temporary root filesystem loaded into memory (RAM) by GRUB before the real root filesystem is mounted. It contains:

- Kernel modules needed to detect and mount the real root filesystem
- Tools like fsck (filesystem check), udev (device manager), cryptsetup (for encrypted drives)
- A minimal BusyBox shell (used for emergency recovery when boot fails)

When initramfs fails to find the root device (e.g., due to a wrong kernel parameter, corrupt filesystem, or missing driver), it drops into a BusyBox emergency shell — which was one of the key errors encountered in this installation.

13.3 What is an EFI System Partition?

The EFI System Partition (ESP) is a small FAT32-formatted partition that stores bootloader files for UEFI-compatible operating systems. On a GPT disk, it is always the first partition and is mounted at /boot/efi in Linux. Each OS installs its bootloader into a subdirectory of the ESP (e.g., /boot/efi/EFI/ubuntu/, /boot/efi/EFI/Microsoft/). The UEFI firmware reads the ESP to determine what to boot.

13.4 What is a Live Environment?

The Ubuntu USB functions as both an installer and a 'Live' operating system. Choosing 'Try Ubuntu' boots a fully functional Ubuntu session running entirely in RAM from the USB drive. No installation is performed. This is useful for:

- Testing hardware compatibility before committing to installation
- Accessing the internet and tools for recovery tasks
- Running GRUB repair commands (grub-install, update-grub) via chroot
- Recovering files from a broken installation

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14. Conclusion

This case study documented a complete, real-world Ubuntu 24.04 LTS installation journey on a modern UEFI-based PC with NVMe storage. The process involved multiple installation attempts due to bootloader conflicts arising from the presence of Windows on the system.

The primary lessons learned from this installation are:

10. Always select 'Erase disk and install Ubuntu' when a clean, single-OS installation is intended. This ensures GRUB is correctly installed and Windows Boot Manager is removed from the UEFI NVRAM.
11. The GPT partition scheme must be selected in Rufus when the target machine uses UEFI firmware (confirmed via msinfo32 → BIOS Mode: UEFI).
12. The initramfs emergency shell is a symptom of a bootloader or partition problem — not a hardware failure. It is recoverable via a Live USB and chroot environment.
13. Third-party drivers should always be selected during installation to ensure Wi-Fi and graphics hardware function correctly out-of-the-box.
14. Post-installation verification using lsblk, df, and cat /etc/os-release provides definitive confirmation of a successful installation.

Ubuntu 24.04 LTS, once correctly installed, provides a stable, well-supported, and fully functional Linux environment suitable for development, research, and general computing. The challenges encountered during this installation are representative of real-world deployment scenarios and demonstrate the importance of understanding UEFI/BIOS configuration, partition schemes, and bootloader mechanics.

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